Class-11th
Chapter- Trigonometric Functions
Hand Written Notes

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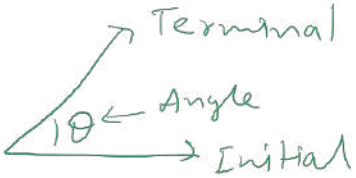
Class-11th

Basic of trigonometry!-

The word trigonometry is derived from three Greek works

Tri → 3 Gon → Angle Metron → Measures

Angle!- The measure of angle is the amount of rotation from the direction of ray of the angle to the another The initial and final position of revolving rays are respectively called the initial side and terminal side



System of Angle Measurement

- English System (Degree)
- French System (grades)
- Circular System (Radians)

Measurement of Angle!-

→ Degree System & Radian (circular system)

→ π radian = 180°

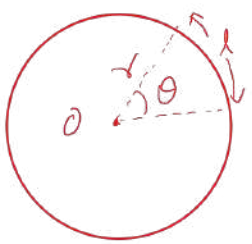
→ 1 radian = $\frac{180^\circ}{\pi}$

1 radian = 57° 16' approximately

→ $1^\circ = \left(\frac{\pi}{180}\right)$ radian

1° = 0.01746 radian approximately

→



θ (in radian) = $\frac{l}{r}$

$r\theta = l$

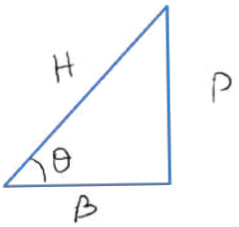
one right angle = 90° degree

1° = 60' (minutes)

1' = 60'' (seconds)

Degree	30°	45°	60°	90°	180°	270°	360°
Radian	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π

$\sin\theta = \frac{P}{H}$, $\cos\theta = \frac{B}{H}$, $\tan\theta = \frac{P}{B} = \frac{\sin\theta}{\cos\theta}$



$\text{cosec}\theta = \frac{H}{P} = \frac{1}{\sin\theta}$, $\sec\theta = \frac{H}{B} = \frac{1}{\cos\theta}$

$\cot\theta = \frac{B}{P} = \frac{\cos\theta}{\sin\theta} = \frac{1}{\tan\theta}$

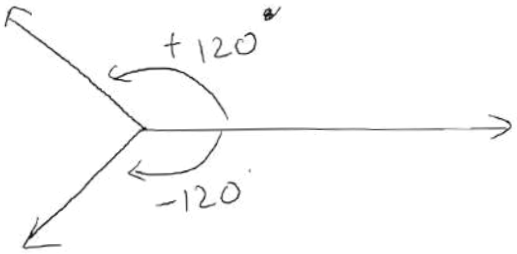
Three basic Identities :-

1. $\sin^2\theta + \cos^2\theta = 1$
2. $\sec^2\theta - \tan^2\theta = 1 \Rightarrow (\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = 1$
3. $\text{cosec}^2\theta - \cot^2\theta = 1 \Rightarrow (\text{cosec}\theta + \cot\theta)(\text{cosec}\theta - \cot\theta) = 1$

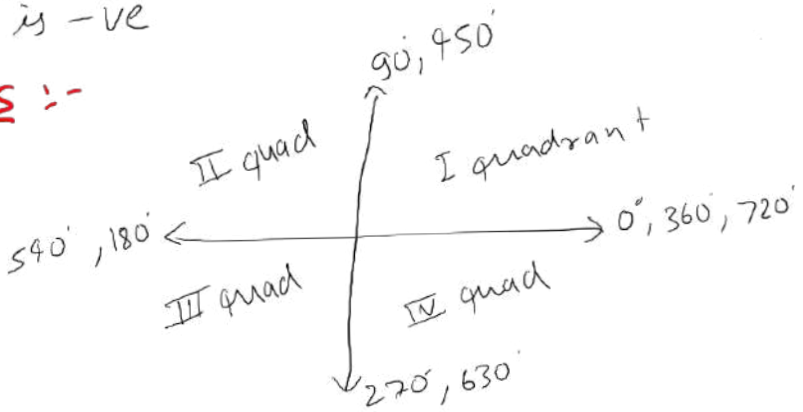
+ve & -ve Angle

→ anticlockwise angle measurement is +ve

→ clockwise angle measurement is -ve



Quadrants :-

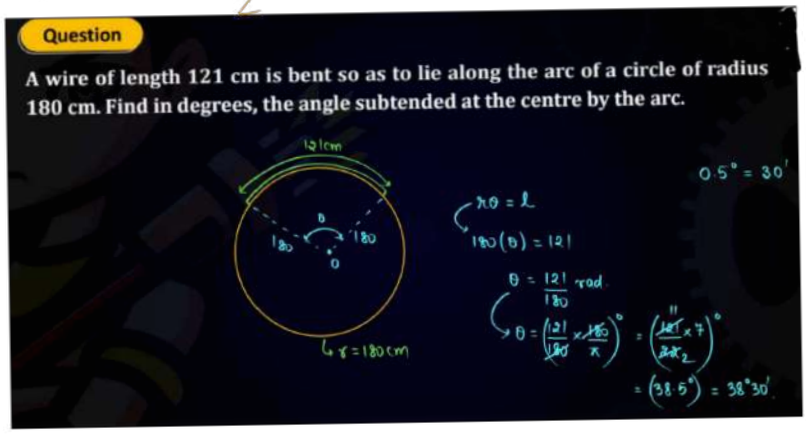


Note :-

i) 90°, 180°, 270°, ⇒ Allied angle

ii) If $\alpha + \beta = \pi \Rightarrow \alpha$ & β supplementary angles

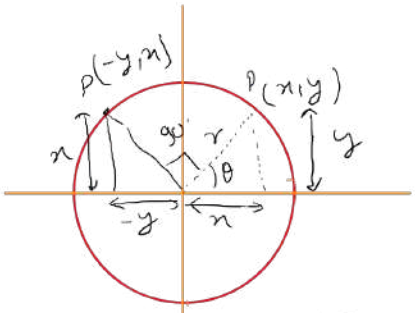
iii) If $\alpha + \beta = \frac{\pi}{2} \Rightarrow \alpha$ & β complementary angles

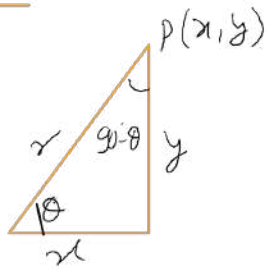


Sine & cosine function :-

$$\sin\theta = \frac{y\text{-coord of point } P}{\text{radius of circle}} = \frac{y}{r}$$

$$\cos\theta = \frac{x\text{-coord of point } P}{\text{radius of circle}} = \frac{x}{r}$$





$\sin(90^\circ-\theta) = \frac{x}{r} = \cos\theta$

$\cos(90^\circ-\theta) = \frac{y}{r} = \sin\theta$

$\sin(90^\circ+\theta) = \frac{x}{r} = \cos\theta$

$\cos(90^\circ+\theta) = \frac{-y}{r} = -\sin\theta$

$\tan(90^\circ+\theta) = -\cot\theta$

$\cot(90^\circ+\theta) = -\tan\theta$

$\sec(90^\circ+\theta) = -\operatorname{cosec}\theta$

$\operatorname{cosec}(90^\circ+\theta) = \sec\theta$

$\tan\theta = \frac{y\text{-cord of point } P}{x\text{-cord of point } P}$

$\tan\theta = \frac{y}{x}$

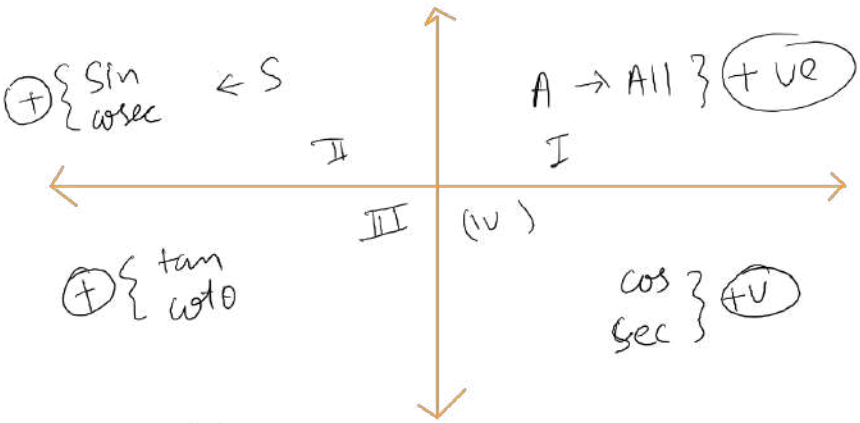
$\tan(90-\theta) = \frac{x}{y} = \cot\theta$

$\cot(90-\theta) = \tan\theta$

$\operatorname{cosec}(90-\theta) = \sec\theta$

$\sec(90-\theta) = \operatorname{cosec}\theta$

Sign of Trigonometric Function in different Quadrants



Property : Dealing with -ve Sign :-

$\sin(-\theta) = -\sin\theta$

$\operatorname{cosec}(-\theta) = -\operatorname{cosec}\theta$

$\tan(-\theta) = -\tan\theta$

$\cot(-\theta) = -\cot\theta$

$\cos(-\theta) = \cos\theta$

$\sec(-\theta) = \sec\theta$

Reduction Formula :-

$\sin\left(\frac{\pi}{2}-\theta\right) = \cos\theta$

$\cos\left(\frac{\pi}{2}-\theta\right) = \sin\theta$

$\tan\left(\frac{\pi}{2}-\theta\right) = \cot\theta$

$\cot\left(\frac{\pi}{2}-\theta\right) = \tan\theta$

$\sec\left(\frac{\pi}{2}-\theta\right) = \operatorname{cosec}\theta$

$\operatorname{cosec}\left(\frac{\pi}{2}-\theta\right) = \sec\theta$

$\sin\left(\frac{\pi}{2}+\theta\right) = \cos\theta$

$\cos\left(\frac{\pi}{2}+\theta\right) = -\sin\theta$

$\tan\left(\frac{\pi}{2}+\theta\right) = -\cot\theta$

$\cot\left(\frac{\pi}{2}+\theta\right) = -\tan\theta$

$\operatorname{cosec}\left(\frac{\pi}{2}+\theta\right) = \sec\theta$

$\sec\left(\frac{\pi}{2}+\theta\right) = -\operatorname{cosec}\theta$

$\sin(\pi-\theta)=\sin\theta$	$\sin(\pi+\theta)=-\sin\theta$
$\cos(\pi-\theta)=-\cos\theta$	$\cos(\pi+\theta)=-\cos\theta$
$\tan(\pi-\theta)=-\tan\theta$	$\tan(\pi+\theta)=\tan\theta$
$\cot(\pi-\theta)=-\cot\theta$	$\cot(\pi+\theta)=\cot\theta$
$\sec(\pi-\theta)=-\sec\theta$	$\sec(\pi+\theta)=-\sec\theta$
$\operatorname{cosec}(\pi-\theta)=\operatorname{cosec}\theta$	$\operatorname{cosec}(\pi+\theta)=-\operatorname{cosec}\theta$

Question

$$\left(\frac{\tan 20^\circ}{\operatorname{cosec}70^\circ}\right)^2+\left(\frac{\cot 20^\circ}{\sec 70^\circ}\right)^2+2\tan15^\circ\tan37^\circ\tan53^\circ\tan60^\circ\tan75^\circ$$

$$\left(\frac{\sin 20^\circ}{\cos 20^\circ}\right)^2+\left(\frac{\cos 20^\circ}{\sin 20^\circ}\right)^2 \quad 2\tan15^\circ\tan37^\circ\tan53^\circ\tan60^\circ\tan75^\circ$$

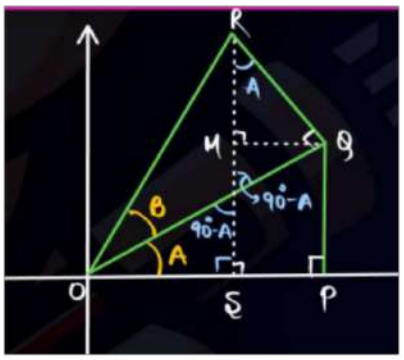
$$\sin^2 20^\circ+\cos^2 20^\circ \qquad \cot(53^\circ)\quad \sqrt{3}\quad \cot(15^\circ)$$

$$1 \qquad 1$$

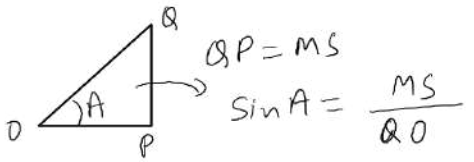
$$1+2\sqrt{3}$$

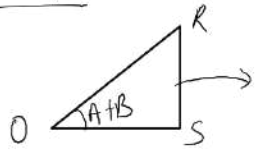
Compound Angle formula:-

I) $\sin(A+B)=\sin A\cos B+\cos A\sin B$



Proof:-





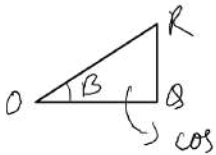
$$\sin(A+B)=\frac{RS}{OR}=\frac{RM+MS}{OR}$$

$$=\frac{RM}{OR}+\frac{MS}{OR}$$

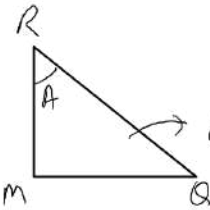
$$=\frac{RM}{QR}\frac{QR}{OR}+\frac{MS}{OQ}\frac{OQ}{OR}$$

$$=\cos A\cdot\sin B+\sin A\cos B$$

$$\sin(A+B)=\sin A\cos B+\cos A\sin B$$



$$\cos B=\frac{OQ}{OR}$$



$$\cos A=\frac{RM}{QR}$$

II) $\sin(A-B)=\sin A\cos B-\cos A\sin B$

III) $\cos(A+B)=\cos A\cos B-\sin A\sin B$

IV) $\cos(A-B)=\cos A\cos B+\sin A\sin B$

V) $\tan(A+B)=\dfrac{\tan A+\tan B}{1-\tan A\tan B}$

VI) $\tan(A-B)=\dfrac{\tan A-\tan B}{1+\tan A\tan B}$

VII) $\cot(A+B)=\dfrac{\cot A\cot B-1}{\cot A+\cot B}$

VII) $\cot(A-B)=\dfrac{\cot A\cot B+1}{\cot A-\cot B}$

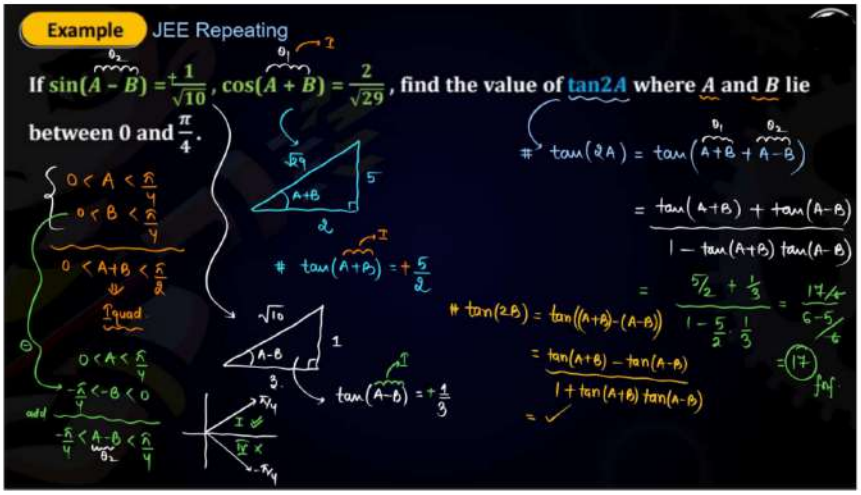
Remember !-

→ $\sin(15^\circ) = \sin\left(\frac{\pi}{12}\right) = \frac{\sqrt{3}-1}{2\sqrt{2}} = \cos(75^\circ) = \cos\left(\frac{5\pi}{12}\right)$

→ $\sin(75^\circ) = \sin\left(\frac{5\pi}{12}\right) = \frac{\sqrt{3}+1}{2\sqrt{2}} = \cos(15^\circ) = \cos\left(\frac{\pi}{12}\right)$

→ $\tan(15^\circ) = \tan\left(\frac{\pi}{12}\right) = 2-\sqrt{3} = \cot(75^\circ) = \cot\left(\frac{5\pi}{12}\right)$

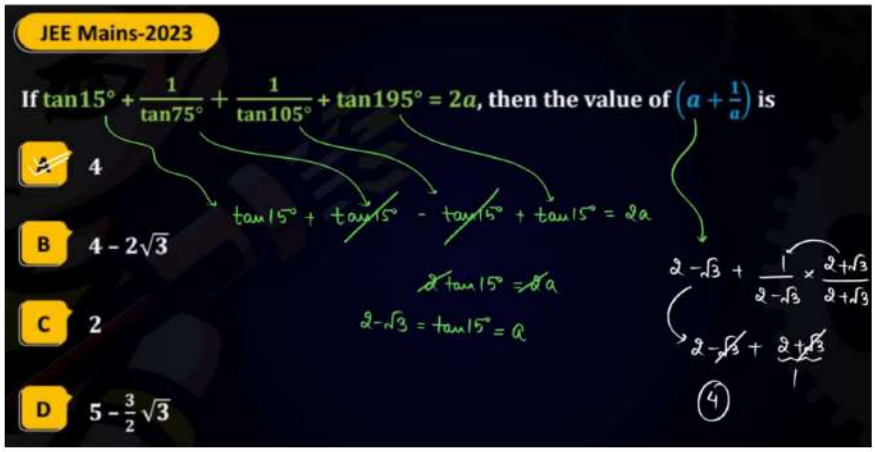
→ $\tan(75^\circ) = \tan\left(\frac{5\pi}{12}\right) = 2+\sqrt{3} = \cot(15^\circ) = \cot\left(\frac{\pi}{12}\right)$



Very important result :-

If $\boxed{A+B = \frac{\pi}{4}}$ then $\boxed{(1+\tan A)(1+\tan B) = 2}$

Important :- $\tan(A+B+C) = \frac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A\tan B - \tan B\tan C - \tan C\tan A}$



Transformation formula :-

I) $\sin C + \sin D = 2\sin\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$

II) $\sin C - \sin D = 2\cos\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$

III) $\cos C + \cos D = 2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$

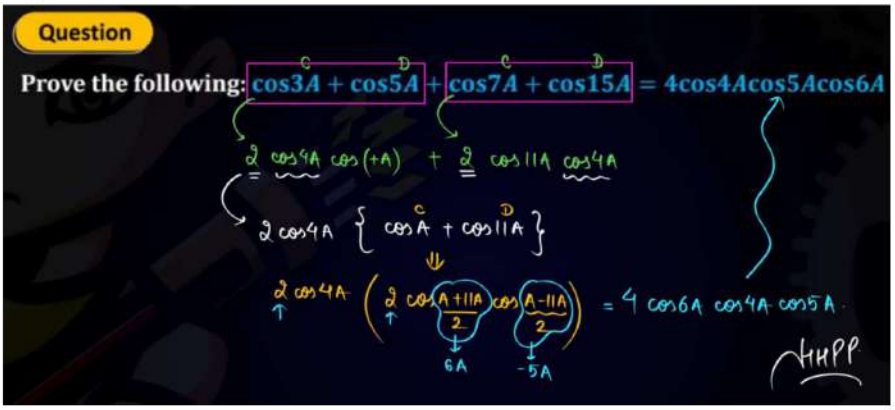
IV) $\cos C - \cos D = -2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$

V) $2\sin A\cos B = \sin(A+B) + \sin(A-B)$

VI) $2\cos A\sin B = \sin(A+B) - \sin(A-B)$

VII) $2\cos A\cos B = \cos(A+B) + \cos(A-B)$

VIII) $2\sin A\sin B = \cos(A-B) - \cos(A-B)$



Formulas for Multiple & Submultiple Angles :-

1) $\sin 2\theta = 2\sin\theta\cos\theta = \frac{2\tan\theta}{1+\tan^2\theta}$

2) $\cos 2\theta = \cos^2\theta - \sin^2\theta$
$= 2\cos^2\theta - 1$
$= 1 - 2\sin^2\theta$
$\Rightarrow \cos 2\theta = \frac{1-\tan^2\theta}{1+\tan^2\theta}$

3) $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta} \Rightarrow \tan^2\theta = \frac{1-\cos 2\theta}{1+\cos 2\theta}$

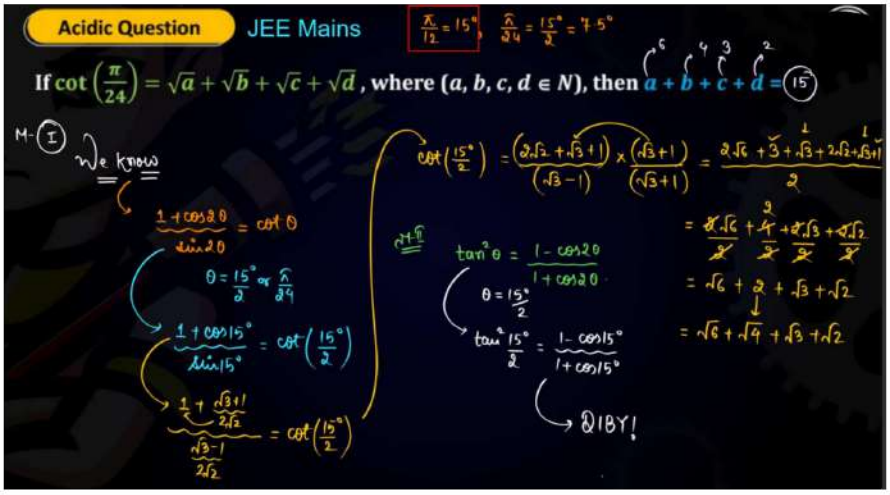
4) $(\sin\theta + \cos\theta)^2 - 1 = \sin 2\theta = 1 - (\sin\theta - \cos\theta)^2$

Note :- 1) $\frac{1+\cos 2\theta}{\sin 2\theta} = \cot\theta$

2) $\frac{1-\cos 2\theta}{\sin 2\theta} = \tan\theta$

3) $\frac{1+\sin 2\theta}{\cos 2\theta} = \tan\left(\frac{\pi}{4}+\theta\right)$

4) $\frac{1-\sin 2\theta}{\cos 2\theta} = \tan\left(\frac{\pi}{4}-\theta\right)$

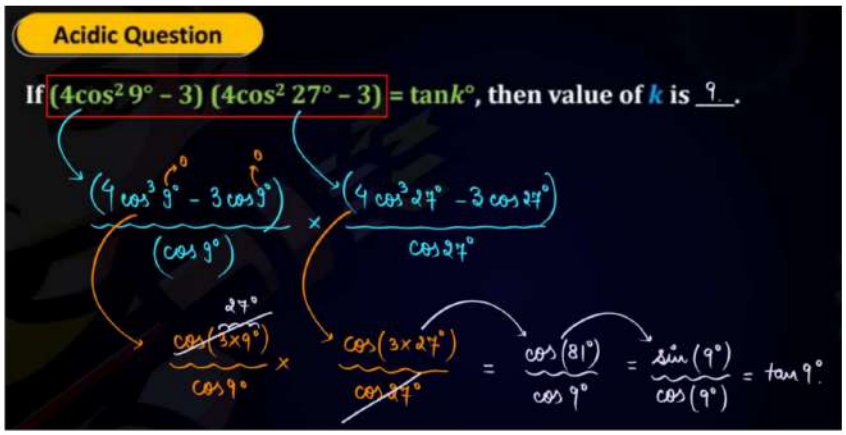


Formulas for Triple Angle :-

1) $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$

2) $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$

3) $\tan 3\theta = \dfrac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$



Important values to Memorise :-

1) $\sin 18^\circ = \dfrac{\sqrt{5}-1}{4} = \cos 72^\circ$

2) $\sin 54^\circ = \dfrac{\sqrt{5}+1}{4} = \cos 36^\circ$

3) $\sin 36^\circ = \dfrac{\sqrt{10-2\sqrt{5}}}{4} = \cos 54^\circ$

4) $\sin 72^\circ = \dfrac{\sqrt{10+2\sqrt{5}}}{4} = \cos 18^\circ$

5) $\tan\left(\frac{\pi}{8}\right) = \sqrt{2}-1 = \cot\left(\frac{3\pi}{8}\right)$

6) $\tan\left(\frac{3\pi}{8}\right) = \sqrt{2}+1 = \cot\left(\frac{\pi}{8}\right)$

#Very Important Result!-

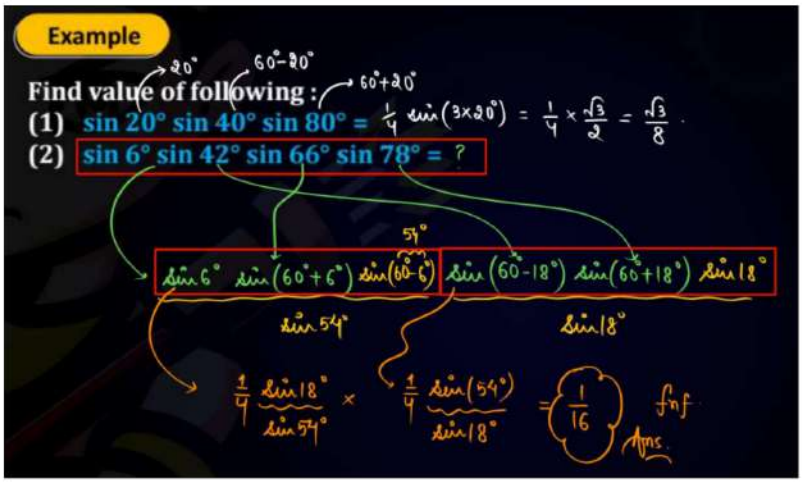
1) $\sin\theta \sin(60^\circ+\theta)\sin(60^\circ-\theta) = \frac{1}{4}\sin 3\theta$

2) $\cos\theta \cos(60^\circ+\theta)\cos(60^\circ-\theta) = \frac{1}{4}\cos 3\theta$

3) $\tan\theta \tan(60^\circ+\theta)\tan(60^\circ-\theta) = \tan 3\theta$

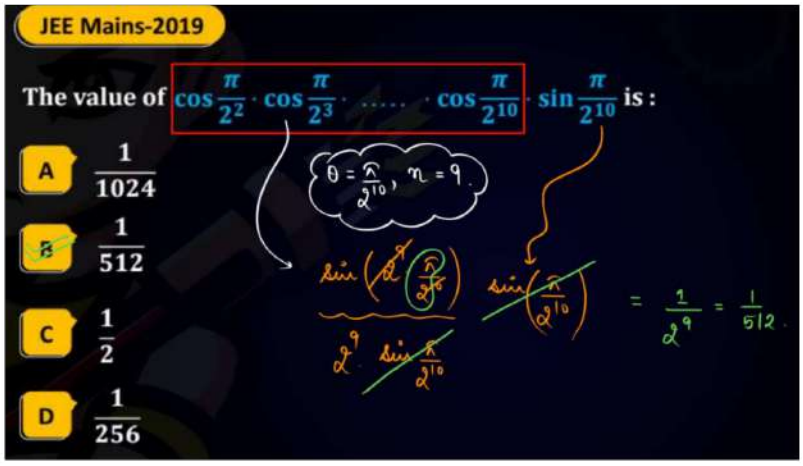
4) $\sin(A+B)\sin(A-B) = \sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A$

5) $\cos(A+B)\cos(A-B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$



Product of cosine with double Angles!-

$$\cos\theta \cdot \cos 2\theta \cdot \cos 4\theta \cdot \cos 8\theta \cdot \dots \cdot \cos(2^{n-1}\theta) = \frac{\sin(2^n\theta)}{2^n \sin\theta}$$



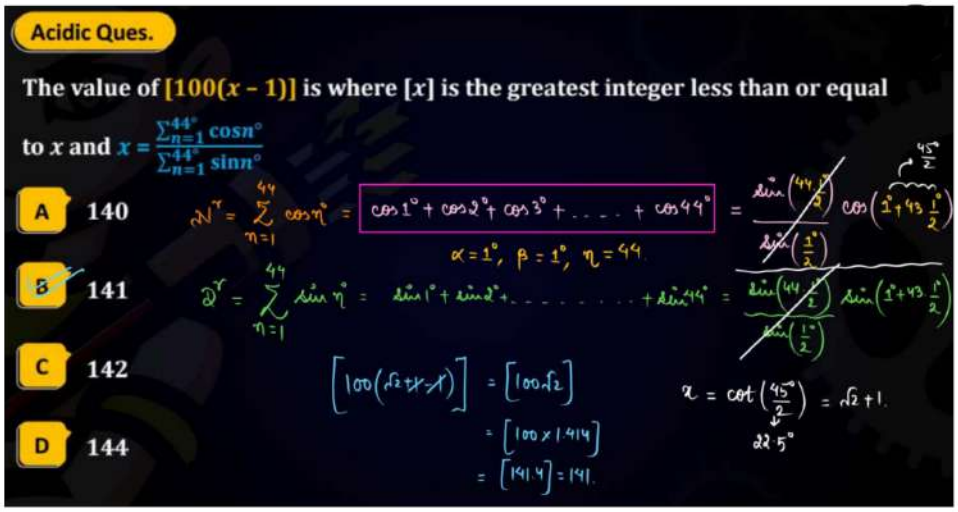
Trigonometric Series!-

Type-01: Sine & cosine series with angle in A.P

a) $\sin\alpha + \sin(\alpha+\beta) + \sin(\alpha+2\beta) + \dots + \sin(\alpha+(n-1)\beta) = \frac{\sin\left(\frac{n\beta}{2}\right)\sin\left(\alpha+(n-1)\frac{\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)}$

b) $\cos\alpha + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + \dots + \cos(\alpha+(n-1)\beta) = \frac{\sin\left(\frac{n\beta}{2}\right)\cos\left(\alpha+(n-1)\frac{\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)}$

$\alpha \to$ first angle, $\beta \to$ common diff of angle, $n \to$ no of terms.

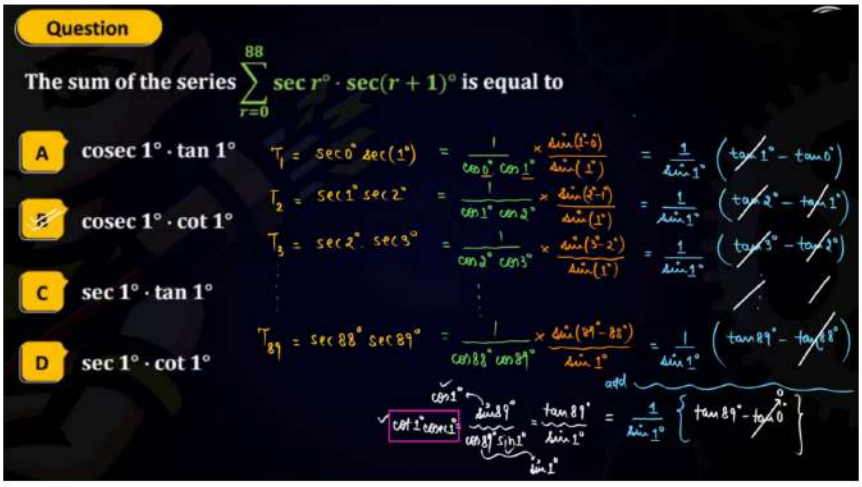


Type -02 : Splitting Tn as difference of two terms

Important Note:- denominator contains product of trigonometric Terms

$$\# \quad \frac{1}{\sin A \sin B} = \frac{\sin(A-B)}{\sin(A-B)\sin A \sin B} = \frac{1}{\sin(A-B)}\left(\frac{\sin A\cos B}{\sin A \sin B} - \frac{\cos A \sin B}{\sin A \sin B}\right) = \frac{\cot B - \cot A}{\sin(A-B)}$$

$$\# \quad \frac{1}{\cos A \cos B} = \frac{\sin(A-B)}{\sin(A-B)\cos A \cos B} = \frac{1}{\sin(A-B)}(\tan A - \tan B)$$



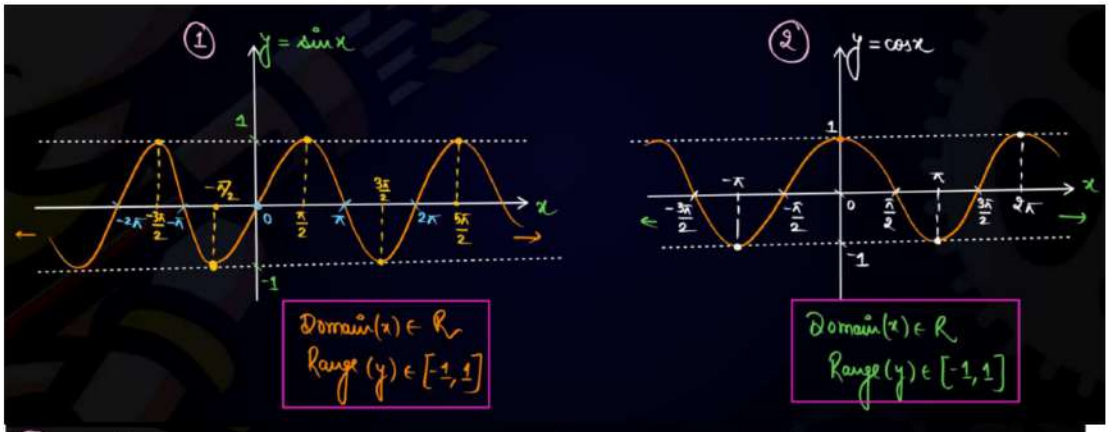
Two important Series Initiations :-

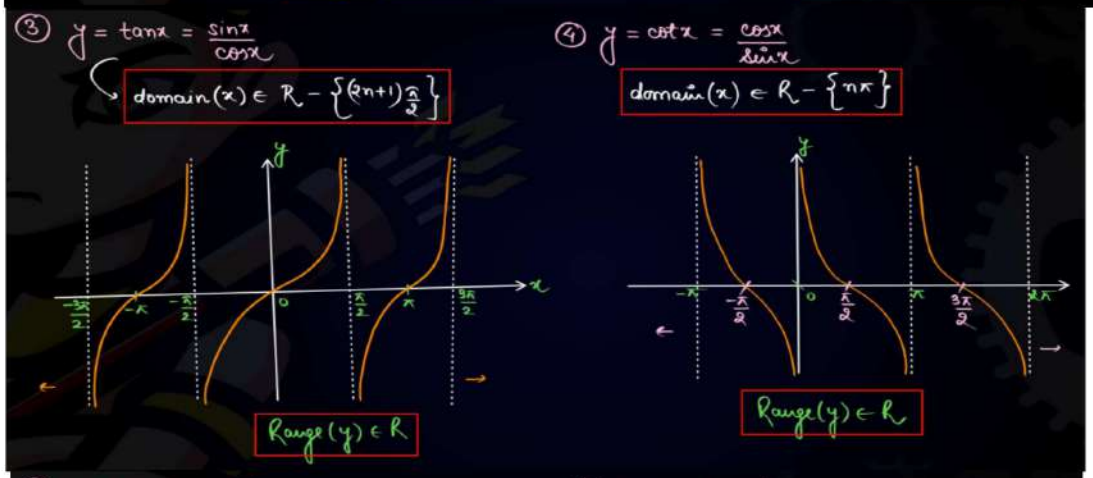
1) $\tan\theta = \cot\theta - 2\cot 2\theta$

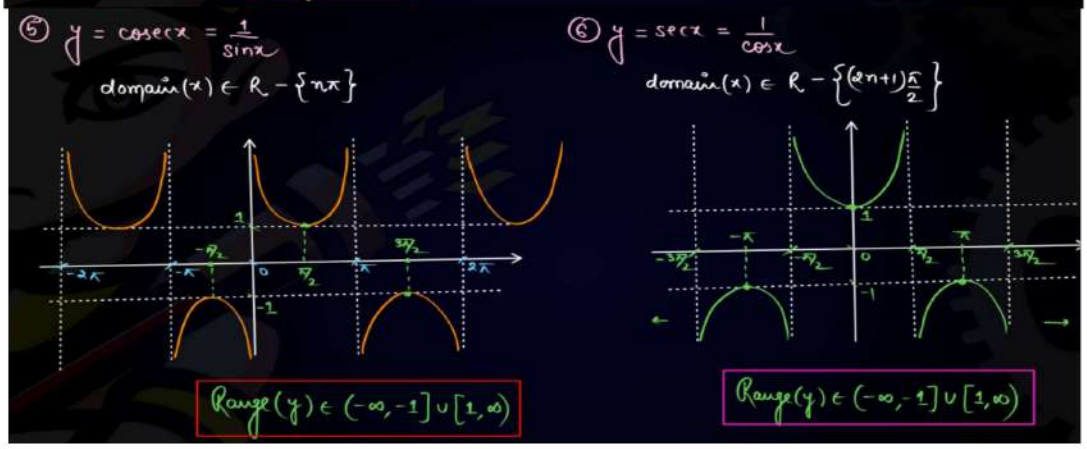
2) $\operatorname{cosec}\theta = \cot\left(\frac{\theta}{2}\right) - \cot\theta$

Graph of Trigonometric Functions :-

Graph, Domain & Range of Six Trigonometric Functions :-







Conditional Identities :-

condition: $A + B + C = \pi$ or A, B & C are angles of triangle

1) $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$

2) $\cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C$

3) $\sin A + \sin B + \sin C = 4 \cos \frac{A}{2} \cos \frac{B}{2} \cos \frac{C}{2}$

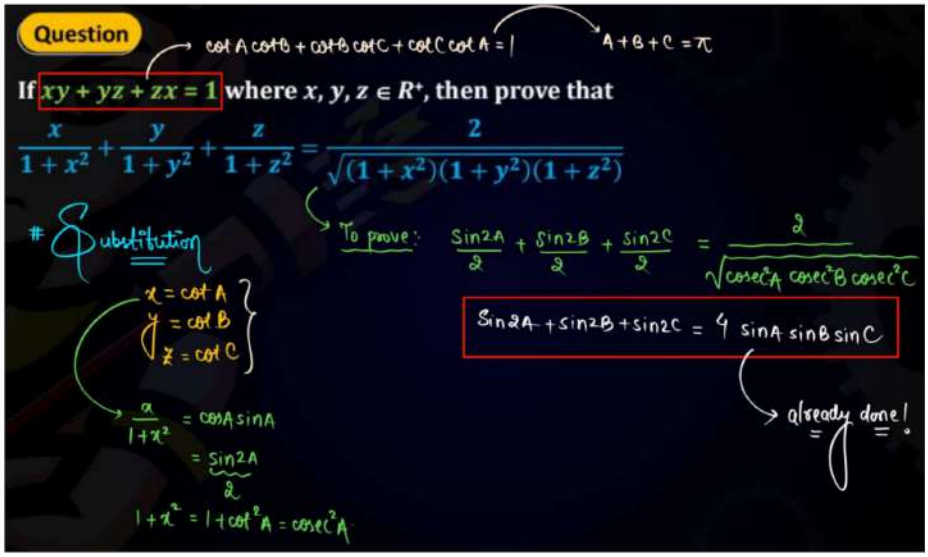
4) $\cos A + \cos B + \cos C = 1 + 4 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}$

5) $\tan A + \tan B + \tan C = \tan A \tan B \tan C$

6) $\cot A \cot B + \cot B \cot C + \cot C \cot A = 1$

7) $\tan 2A + \tan 2B + \tan 2C = \tan 2A \tan 2B \tan 2C$

8) $\tan \frac{A}{2} \tan \frac{B}{2} + \tan \frac{B}{2} \tan \frac{C}{2} + \tan \frac{C}{2} \tan \frac{A}{2} = 1$



Range of Trigonometric Functions :-

Type -01 :- Based on Range of Standard Trigonometric Functions :-

$\sin x \in [-1, 1]$

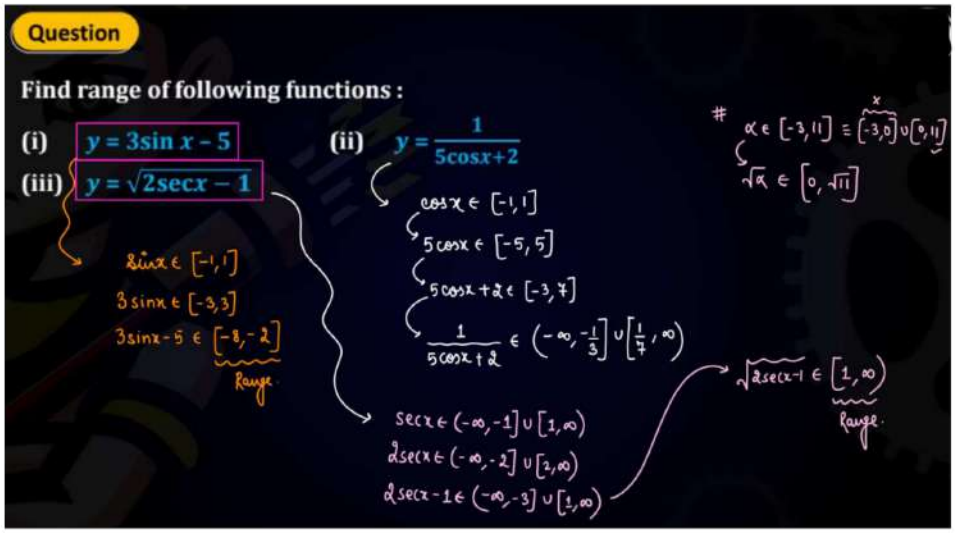
$\cos x \in [-1, 1]$

$\tan x \in R$

$\cot x \in R$

$\sec x \in (-\infty, -1] \cup [1, \infty)$

$\operatorname{cosec} x \in (-\infty, -1] \cup [1, \infty)$



Type-02: Range by completing perfect square :-

Ex → 1) $y = \cos^2 x - 4\cos x + 13$

$y = \cos^2 x - 4\cos x + 4 + 9 \Rightarrow$

$y = (\cos x - 2)^2 + 9$

$y \in [10, 18]$

$-1 \le \cos x \le 1$

$-3 \le \cos x - 2 \le -1$

$1 \le (\cos x - 2)^2 \le 9$

$10 \le (\cos x - 2)^2 + 9 \le 18$

Type-03: Range of Trigonometric Functions

form :- $f(x) = a\cos x \pm b\sin x$

Range $\in \left[-\sqrt{a^2+b^2}, \sqrt{a^2+b^2}\right]$

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Find the maximum & minimum value of :

$$y = \frac{1}{\sin^2\theta + 3\sin\theta\cos\theta + 5\cos^2\theta} = \frac{1}{f(\theta)}$$

$$f(\theta) = \sin^2\theta + 3\sin\theta\cos\theta + 5\cos^2\theta$$

$$f(\theta) = \left(\frac{1-\cos 2\theta}{2}\right) + 3\,\frac{\sin 2\theta}{2} + 5\left(\frac{1+\cos 2\theta}{2}\right)$$

$$f(\theta) = \frac{1}{2} - \frac{1}{2}\cos 2\theta + \frac{3}{2}\sin 2\theta + \frac{5}{2} + \frac{5}{2}\cos 2\theta$$

$$= \frac{2\cos 2\theta + \frac{3}{2}\sin 2\theta}{} + 3$$

$$\sqrt{2^2 + \left(\frac{3}{2}\right)^2} = \sqrt{4 + \frac{9}{4}} = \frac{5}{2} \rightarrow \left[-\frac{5}{2}, \frac{5}{2}\right] + 3 \Rightarrow \left[\frac{1}{2}, \frac{11}{2}\right]$$

$$f(\theta) \in \left[\frac{1}{2}, \frac{11}{2}\right]$$

$$\frac{1}{f(\theta)} \in \left[\frac{2}{11}, 2\right]$$

Type-04 Special form :-

i) $a^2\tan^2\theta + b^2\cot^2\theta \geq 2ab$

ii) $a^2\sec^2\theta + b^2\operatorname{cosec}^2\theta \geq (a+b)^2$

